

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. Let $T : \mathcal{P}^3(\mathbb{R}) \rightarrow \mathbb{R}$ be a linear transformation for which

$$T(1) = 1, \quad T(1 + t) = 2, \quad T(1 + t + t^2) = 3, \quad T(1 + t + t^2 + t^3) = 4.$$

- (a) Find $T(1 + 3t + 4t^2 + 2t^3)$.
(b) Find $T(a + bt + ct^2 + dt^3)$ for any $a, b, c, d \in \mathbb{R}$.

2. Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be a linear transformation for which

$$T(a, b, c, d) = (a + b, b + c, c + d, d + a)$$

- (a) Find a basis for $N(T)$.
(b) Find a basis for $R(T)$.

3. Prove the following parts of Theorem 2.26:

- (a) (1) T is injective if and only if $\text{rank}(T) = \dim(V)$.
(b) (5) If $\dim(V) < \dim(W)$, then T is not surjective.

4. Define the following complex vector space

$$V = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{C}^{2 \times 2} : a + d = 0 \right\}$$

with the usual addition and scalar multiplication.

- (a) Find an integer r such that $V \simeq \mathbb{C}^r$.
(b) Give an explicit isomorphism $T : V \rightarrow \mathbb{C}^r$. Make sure to prove that T is an isomorphism!

5. Let $A, B \in \mathbb{F}^{n \times n}$ be invertible matrices.

- (a) Prove that AB is invertible with $(AB)^{-1} = B^{-1}A^{-1}$.
(b) Prove that A^T is invertible with $(A^T)^{-1} = (A^{-1})^T$.



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 2.1: 1, 5, 6, 7, 11, 13
- Section 2.2: 1, 2, 4, 5, 9, 11, 13
- Section 2.3: 1, 4, 5, 9, 10, 11
- Section 2.7: 2, 13, 19, 24

1. Prove NS Theorem 2.33 (3): Let $T : V \rightarrow W$ and $S : U \rightarrow V$ be isomorphisms. Prove that $TS : U \rightarrow W$ is an isomorphism and $(TS)^{-1} = S^{-1}T^{-1}$.
2. Give an example of a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $R(T) = N(T)$.
3. Give an example of vector spaces V and W and distinct linear transformations $T_1 : V \rightarrow W$ and $T_2 : V \rightarrow W$ such that $N(T_1) = N(T_2)$ and $R(T_1) = R(T_2)$.

